

COURSE SYLLABUS

Introduction to SQL Databases (2025)

COURSE NUMBER

55315B

DURATION

3 days · 9 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This three-day instructor-led course is aimed at people looking to move into a database professional role or whose job role is expanding to encompass database elements. The course describes fundamental database concepts including database types, database languages, and database designs, with a focus on SQL Server 2025 and SSMS 22. Students gain hands-on experience exploring, querying, designing, and managing SQL Server databases using industry-standard techniques.

AT COURSE COMPLETION

After completing this course, you will be able to:

Describe and apply key relational database concepts in SQL Server 2025, including database types, the ANSI/SPARC architecture, SQL language categories, and new SQL Server 2025 and SSMS 22 features

Design efficient relational database schemas by applying entity relationship diagramming, normalization rules through Third Normal Form, and selective denormalization strategies

Implement and enforce data integrity across related tables in SQL Server 2025 using foreign key constraints with cascading options, DML triggers, and DDL triggers

Evaluate and improve database performance by designing effective indexing strategies, rewriting nonsargable queries, and understanding the effects of transaction isolation levels and concurrency control

Create and use essential SQL Server 2025 database objects — including tables with constraints, standard and indexed views, stored procedures with parameters, DML and DDL triggers, and scalar and table-valued user-defined functions — to meet the needs of database administrators and application users

PREREQUISITES

- General computer literacy and basic experience working with personal computers

- No prior database experience required
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COURSE OUTLINE

MODULE 1 Introduction to Databases

147 min

This module introduces key database concepts in the context of SQL Server 2025, covering relational database fundamentals, alternative storage types, data analysis concepts, and the SQL language categories used in SQL Server. Students explore the new features in SQL Server 2025 and SSMS 22 and practice querying tables using T-SQL SELECT statements.

LESSONS

Lesson 1: What's New in SQL Server 2025 and SSMS 22

Lesson 2: Introducing Relational Databases

Lesson 3: Other Database Types: OLAP, NoSQL, and Data Warehouses

Lesson 4: Data Analysis Concepts: From Raw Data to Intelligence

Lesson 5: SQL Server Database Languages: DDL, DML, DCL, TCL, and DQL

Lesson 6: Writing T-SQL SELECT Statements

LAB EXERCISES

Exploring SQL Server 2025 Databases and Tables

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Describe the structure and purpose of relational databases, including tables, rows, columns, and keys, in the context of SQL Server 2025

Differentiate between relational (OLTP), OLAP, and NoSQL database types and identify appropriate use cases for each

Explain the data analytics pipeline from raw data collection through interpretation to actionable intelligence

Identify and use the five SQL language categories — DDL, DML, DCL, TCL, and DQL — supported by SQL Server 2025

Describe and apply new features introduced in SQL Server 2025 and SSMS 22

Write basic T-SQL SELECT statements with WHERE clause filtering to retrieve targeted data from SQL Server 2025 tables

This module describes data modeling techniques used in relational database design, including Entity Relationship Diagramming, the ANSI/SPARC three-schema architecture, and relationship modeling with primary keys, foreign keys, and junction tables. Students apply these techniques to a real-world scenario.

LESSONS

Lesson 1: Introduction to Data
Modeling and ERDs

Lesson 2: The ANSI/SPARC Three-
Schema Architecture

Lesson 3: Identifying Entities,
Attributes, and Primary Keys

Lesson 4: Modeling Relationship Types:
One-to-One, One-to-Many,
and Many-to-Many

Lesson 5: Finalizing a Data Model:
Constraints, Assumptions,
and Documentation

LAB EXERCISES

Modeling a Database for a Course
Registration System

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Explain the purpose and components of an Entity Relationship Diagram (ERD) including entities, attributes, and relationships

Describe the three levels of the ANSI/SPARC architecture — external, conceptual, and internal — and explain how each level serves different stakeholders

Identify primary keys, composite keys, and candidate keys to uniquely identify entities in a data model

Model one-to-one, one-to-many, and many-to-many relationships and resolve many-to-many relationships using junction tables

Apply crow's foot notation to represent cardinality in an ERD

Design a logical data model for a given business scenario by identifying entities, attributes, relationships, and constraints

This module describes normalization and denormalization techniques, covering the purpose and process of applying First, Second, and Third Normal Form rules to relational database schemas, and when to selectively denormalize for performance. Students normalize an unnormalized data set through 3NF using SQL Server 2025.

LESSONS

- Lesson 1:** Fundamentals of Normalization: Goals, Keys, and Dependencies
- Lesson 2:** First Normal Form: Atomic Values and Primary Key Identification
- Lesson 3:** Second Normal Form: Eliminating Partial Dependencies
- Lesson 4:** Third Normal Form and BCNF: Removing Transitive Dependencies
- Lesson 5:** Denormalization Strategies and Their Trade-offs

LAB EXERCISES

- Normalizing Data in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain the benefits of normalization including reduced data duplication, improved insert and update performance, and elimination of update anomalies
- Identify candidate keys, composite primary keys, and surrogate keys in a table design
- Apply First Normal Form by eliminating repeating groups and establishing a primary key
- Apply Second Normal Form by removing partial dependencies from composite-key tables
- Apply Third Normal Form by removing transitive dependencies between non-key attributes
- Describe when and how to selectively denormalize a schema to improve query and reporting performance, and explain the trade-offs involved

This module describes relationship types and their effects in database design, covering how to implement and test foreign key constraints, declarative and procedural referential integrity, cascading options, and triggers in SQL Server 2025 to maintain data consistency across related tables.

LESSONS

- Lesson 1:** Relationship Types in SQL Server 2025: One-to-One, One-to-Many, and Many-to-Many
 - Lesson 2:** Declarative Referential Integrity: Foreign Key Constraints and Their Options
 - Lesson 3:** Procedural Referential Integrity: AFTER and INSTEAD OF Triggers
 - Lesson 4:** Planning a Referential Integrity Strategy
 - Lesson 5:** Testing and Validating Referential Integrity in SSMS
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LAB EXERCISES

- Planning and Implementing Referential Integrity in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe one-to-one, one-to-many, and many-to-many relationship implementations in SQL Server 2025 using primary and foreign keys
- Differentiate between declarative referential integrity using foreign key constraints and procedural referential integrity using triggers
- Implement foreign key constraints with NO ACTION, CASCADE, SET NULL, and SET DEFAULT options for both DELETE and UPDATE operations
- Test referential integrity by writing T-SQL INSERT, UPDATE, and DELETE statements that both satisfy and violate defined constraints
- Create AFTER and INSTEAD OF DML triggers to enforce complex business rules that go beyond declarative foreign key constraints
- Plan and document a referential integrity strategy for a multi-table relational schema

This module introduces the effects of database design decisions on performance in SQL Server 2025, covering indexing strategies, sargable vs. nonsargable query patterns, join operation types, and the effects of concurrency control and transaction isolation levels on throughput and data consistency.

LESSONS

Lesson 1: Clustered and Nonclustered Indexes: Structure and Selection Strategy

Lesson 2: Sargable vs. Nonsargable Queries: Writing Index-Friendly T-SQL

Lesson 3: Join Types and Query Optimizer Behavior

Lesson 4: ACID Properties and Concurrency Problems

Lesson 5: Transaction Isolation Levels, Locking, and the Enhanced Query Store

LAB EXERCISES

Exploring Index Performance, Query Optimization, and Concurrency in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Describe the structure and purpose of clustered and nonclustered indexes in SQL Server 2025 and select appropriate index strategies for given query patterns

Identify nonsargable query patterns that prevent index utilization and rewrite them as sargable alternatives

Explain how the SQL Server 2025 query optimizer selects join operation types — hash, loop, and merge — based on table size, indexes, and data distribution

Describe ACID transaction properties and identify the concurrency problems — dirty reads, nonrepeatable reads, phantom reads, and lost updates — that arise when these properties are compromised

Explain how transaction isolation levels and locking interact to control concurrent data access in SQL Server 2025

Evaluate index design decisions using execution statistics and the enhanced Query Store in SSMS 22

MODULE 6 SQL Server 2025 Database Objects: Tables and Constraints

130 min

This module covers the creation and management of SQL Server 2025 tables with the full range of data types, NULL constraints, primary keys, unique keys, foreign keys, check constraints, default constraints, identity columns, and schemas. Students create and test a complete set of table objects using T-SQL in SSMS 22.

LESSONS

Lesson 1: SQL Server 2025 Data Types: Choosing the Right Type for Each Column

Lesson 2: Table Constraints: Primary Keys, Unique Keys, NOT NULL, and Identity Columns

Lesson 3: Enforcing Business Rules with CHECK and DEFAULT Constraints

Lesson 4: Database Schemas and SQL Server 2025 Ledger Tables

LAB EXERCISES

SQL Server 2025 Database Objects: Tables and Constraints

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Select appropriate SQL Server 2025 numeric, character, date/time, binary, and spatial data types for column definitions

Create tables with NOT NULL constraints, primary keys, unique key constraints, foreign key constraints, check constraints, and default constraints using T-SQL

Configure identity columns with seed and increment values to generate surrogate primary keys automatically

Create and use database schemas to logically organize and secure database objects in SQL Server 2025

Test each constraint type by writing DML statements designed to both satisfy and violate the constraint

Use new SQL Server 2025 Ledger table features to create tamper-evident append-only and updatable ledger tables

This module covers the creation and use of standard views and indexed views in SQL Server 2025, including how views simplify data access, enforce column-level security, and how indexed views with SCHEMABINDING improve performance for mission-critical queries. Students create, query, and manage views using T-SQL in SSMS 22.

LESSONS

- Lesson 1:** What Views Are and Why They Are Used
- Lesson 2:** Creating Standard Views: Syntax, Multi-Table Joins, and Derived Columns
- Lesson 3:** Securing Data Access Through View Permissions
- Lesson 4:** Indexed Views: SCHEMABINDING, Clustered Indexes, and Performance Trade-offs
- Lesson 5:** Managing Views in SSMS 22: Inspection, Dependencies, and Modifications

LAB EXERCISES

- Creating and Managing Views and Indexed Views in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe the purpose of views as stored queries that simplify data access and provide column-level security for end users
- Create standard views using CREATE VIEW with SELECT statements that span multiple tables and include derived columns
- Grant permissions on views to provide end-user access to underlying data without exposing the base table structure
- Create indexed views using the SCHEMABINDING option and a unique clustered index to persist view results for improved query performance
- Explain the maintenance implications of indexed views when source tables are modified
- Use new SSMS 22 features to inspect and manage view definitions and dependencies in Object Explorer

This module covers the creation and use of stored procedures with input parameters and EXECUTE AS, and DML and DDL triggers in SQL Server 2025. Students create, test, and manage these programmability objects using T-SQL in SSMS 22, applying them to automate complex tasks and enforce business rules.

LESSONS

Lesson 1: Stored Procedures: Purpose, Structure, and Pre-Compilation Benefits

Lesson 2: Input Parameters and EXECUTE AS: Flexibility and Indirect Permission Grants

Lesson 3: DML Triggers: AFTER and INSTEAD OF Trigger Types and the inserted/deleted Tables

Lesson 4: DDL Triggers and Logon Triggers: Governance and Security Automation

Lesson 5: Managing and Testing Stored Procedures and Triggers in SSMS 22

LAB EXERCISES

Creating and Using Stored Procedures and Triggers in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Create stored procedures with and without input parameters using T-SQL in SQL Server 2025

Use the EXECUTE AS option in stored procedures to indirectly grant users permission to perform DML operations

Execute stored procedures using EXEC and EXECUTE statements with named and positional parameter syntax

Create AFTER DML triggers that respond to INSERT, UPDATE, and DELETE operations to enforce business rules and maintain audit records

Create INSTEAD OF DML triggers that replace standard table operations with custom logic

Create DDL triggers at the database and server levels to respond to schema change events and enforce governance policies

MODULE 9 User-Defined Functions and Advanced SQL Server 2025 Features

131 min

This module covers the creation and use of scalar, inline table-valued, and multi-statement table-valued user-defined functions in SQL Server 2025, and introduces advanced platform features including enhanced JSON support, temporal tables, and AI-assisted query capabilities in SSMS 22. Students build and test all three function types and explore the new features through guided exercises.

LESSONS

- Lesson 1:** Scalar User-Defined Functions: Syntax, Return Values, and Use in Expressions
- Lesson 2:** Table-Valued Functions: Inline vs. Multi-Statement
- Lesson 3:** Functions vs. Stored Procedures: Choosing the Right Programmability Object
- Lesson 4:** SQL Server 2025 Temporal Tables: Querying Point-in-Time Historical Data
- Lesson 5:** SQL Server 2025 JSON Enhancements and SSMS 22 AI-Assisted Query Features

LAB EXERCISES

- User-Defined Functions and Advanced SQL Server 2025 Features

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and execute scalar user-defined functions that accept input parameters and return a single value for use in T-SQL expressions
- Create and query inline table-valued functions that return a table result set based on a parameterized SELECT statement
- Create and query multi-statement table-valued functions that build and return a result set using multiple T-SQL statements
- Differentiate between stored procedures and user-defined functions in terms of return values, usage in SELECT statements, and side-effect restrictions
- Use SQL Server 2025 temporal table features to query point-in-time historical data
- Apply SSMS 22 AI-assisted query suggestions and enhanced JSON functions to practical T-SQL development scenarios

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3

SKILLS TO BE INTEGRATED

Connect to a SQL Server 2025 instance using SSMS 22 and explore database objects including schemas, tables, and database diagrams

Write and execute T-SQL SELECT statements with WHERE clauses, column filtering, and range conditions to retrieve targeted result sets

Identify entities, attributes, and relationships from a business scenario and produce an Entity Relationship Diagram with correctly resolved many-to-many relationships

Apply First Normal Form, Second Normal Form, and Third Normal Form rules to an unnormalized data set, decomposing tables to eliminate redundancy and transitive dependencies

Selectively denormalize a normalized schema to support anticipated query patterns and explain the trade-offs involved

DAY 2 CAPSTONE Modules 4, 5, 6

SKILLS TO BE INTEGRATED

Create foreign key constraints between related tables using T-SQL in SSMS 22, specifying NO ACTION, CASCADE, SET NULL, or SET DEFAULT referential actions for DELETE and UPDATE operations

Test referential integrity by attempting INSERT, UPDATE, and DELETE operations that both satisfy and violate defined foreign key constraints

Implement INSTEAD OF and AFTER triggers to enforce complex referential integrity rules that go beyond declarative foreign key constraints

Design and create clustered and nonclustered indexes on appropriate columns based on anticipated query patterns

Identify nonsargable query patterns and rewrite queries to be sargable, comparing execution statistics before and after index use in SSMS 22

Describe and demonstrate the effects of transaction isolation levels and locking on concurrent database operations

DAY 3 CAPSTONE Modules 7, 8, 9

SKILLS TO BE INTEGRATED

Create tables with appropriate data types, NULL/NOT NULL constraints, primary keys, unique constraints, check constraints, default constraints, and identity columns using T-SQL in SSMS 22

Create standard views and indexed views with SCHEMABINDING to provide simplified and secured data access for end users

Write and execute stored procedures with input parameters using EXECUTE AS to indirectly grant users access to DML operations

Create DML triggers (AFTER and INSTEAD OF) and DDL triggers to enforce business rules and audit schema changes

Create scalar and table-valued user-defined functions and use them within SELECT statements

Use new SQL Server 2025 features and SSMS 22 enhancements — including enhanced Query Store, Ledger tables, and AI-assisted query suggestions — to create and manage database objects

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.